



Physics-guided deep learning for real-time resonance diagnostics and predictive maintenance in pulsed water jet manufacturing systems

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Abstract

Unlike closure mechanisms for laser or EDM machining, pulsed water jet manufacturing does not currently have any. The reason is timing; nothing tracks the chamber resonance quickly enough. Although FFT analysis requires 30–50 ms, control of a HAPOS (hydrodynamic acoustic pulse oscillatory system) operation up to 450 MPa has to be kept below 10 ms of inference budget. The answer is our 12-layer physics-guided CNN (PG-CNN) which reads the resonance frequency f_{rez} directly from a model's raw pressure signals (100 kHz, Hann-windowed, $N = 4096$ samples). Training runs in two stages. First we pre-train on 2400 synthetic spectra from a validated Helmholtz model ($\Delta f = 6.4\%$ against the five-element HAPOS); we then fine-tune on 129 quality-screened experimental spectra (out of 312 planned acquisition slots) over 5–25 MPa, using discriminative learning rates and early stopping. The dilated convolutions (kernel $K = 7$, dilations $\{1, 2, 4\}$) cover a 1.69 ms receptive field; attention pooling then selects the time segments. Within PG-SAD, the fixed transfer constant $\tilde{k}_{rkk} = 9.41 \cdot 10^{-4}$ gives way to a learnable function $k_{rkk}(p, g)$ under heteroscedastic constraints. On the held-out set ($n = 240$) the PG-CNN reaches 98.2% accuracy inside a ± 50 Hz band with MAE = 23.4 Hz ($R^2 = 0.998$), while PG-SAD lowers the mean absolute frequency error to 12.8 Hz at F1 = 0.97. After ONNX/TensorRT optimisation, a Jetson Xavier NX delivers 6.2 ms inference (161 FPS at 15 W), well within the $\tau \approx 50$ ms time constant of plunger pumps. This physics–CNN bridge is what supplies the sensing layer that closed-loop control and predictive maintenance of pulsed water jet cells have been missing.

Keywords Intelligent manufacturing · Physics-guided deep learning · Real-time process monitoring · Pulsed water jet cutting · Predictive maintenance · Convolutional neural network

List of symbols

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f_{rez}	Resonance frequency of forced steady oscillations [Hz]
f_{ZM}	Experimentally measured frequency [Hz]
f_{VYP}	Indirectly measured (calculated) frequency [Hz]
m_a	Total acoustic mass of the system [kg m ⁻⁴]
m_{aP}, m_{aH}, m_{aT}	Partial acoustic masses of elements, i.e. inlet pipe, throat, nozzles
c_a	Total acoustic compliance of the system [m ⁴ s ² kg ⁻¹]
c_{aUK}, c_{aRK}	Partial acoustic compliance of elements, i.e. inlet and resonance chambers

r_a	Total acoustic resistance of the system [kg m ⁻⁴ s ⁻¹]
k_{rkk}	Transfer (relative correction correlation) coefficient [–]
p_{CERP}	Effective pressure of the liquid by the pump [Pa]
ρ	Density of compressed fluid [kg m ⁻³]
v	Phase velocity of the wave in the elements [m s ⁻¹]
Re	Reynolds number [–]
S	Cross-sectional area of the element [m ²]
l	Element length [m]
V	Resonance chamber volume [m ³]
$\tilde{k}_{rkk}$	Baseline transfer coefficient of the analytical Helmholtz model
$k_{rkk}^{CNN}(p, g)$	Pressure- and geometry-dependent transfer coefficient learned by the CNN
$\hat{f}_{rez}$	Network (PG-CNN) estimate of the resonance frequency
$\Phi(p)$	Dimensionless validity index of the resonance-frequency extrapolation

Introduction

Pulsed water jet cutting is steadily finding its place in modern manufacturing. Applications run from stripping coatings off turbine blades to the selective demolition of civil structures. What it still lacks, unlike laser or EDM machining, is intelligent closed-loop control. The root of the problem is the sensing layer. Resonance inside the cutting head shifts with several variables, chiefly pump pressure, nozzle geometry and early-stage wear. Conventional spectral analysis is simply too slow and too brittle to drive a real-time controller. As factories move toward Industry 4.0, building machine-learning diagnostics into the process-control architecture has turned from a research curiosity into an engineering necessity. The research discussed in this paper closes this gap by developing a physics-based deep learning model, which will be executed on a low-power edge device that is located within an industrial water jet system to provide two forms of predictive maintenance: real-time monitoring of the resonance and early anomaly warning. Compared with a continuous jet, a pulsed configuration delivers higher cutting efficiency (Hlaváč et al., 2009). The pulsing is generated by a hydrodynamic acoustic pulse oscillatory system (HAPOS), which uses acoustic resonance in multiple tubes and a resonance chamber to generate kilohertz frequency oscillations, functioning at 5–450 MPa of pressure (Hlaváč et al., 2014). A chamber system's resonance frequency f_{rez} is set by three acoustic quantities: its mass m_a , its compliance c_a and its

resistive loss r_a . Therefore, finding the correct resonance frequency in real time is not optional—closed-loop control of cutting parameters depends on it (Ward et al., 2021). The traditional route to a frequency spectrum is the fast Fourier transform (FFT), which runs a peak-detecting algorithm over a windowed slice of the signal (Oppenheim & Schaffer, 2010). That took roughly 30–50 ms in total, far too slow to control the HAPOS at kHz-range frequencies. Worse still, the FFT must separate the true resonance peaks from harmonic artefacts and from turbulent noise at high Reynolds numbers (of order 10⁵ across the calibrated 5–25 MPa range, and approaching 1.4·10⁶ near 450 MPa) (Brunton et al., 2020). Neural networks sidestep both problems at once: they learn spectral patterns straight from raw, noisy signals, so no hand-designed feature extraction is needed. Recent reviews of deep learning in spectral analysis confirm this advantage across diverse spectroscopic and signal-processing tasks (Liu et al., 2024).

Our PG-CNN is meant to strengthen the standard spectral techniques (FFT, MUSIC, CWT), not to replace them. At a latency of about 45 ms, FFT-based peak detection cannot feed a real-time closed-loop controller for a HAPOS oscillating in the kHz range; the CNN inference path, by contrast, needs 6.2 ms (TensorRT-optimised Jetson Xavier NX, and 8.3 ms on the RTX 3080 reference GPU, Table 7). In deployment we keep the FFT running alongside, for PG-SAD parallel diagnostics and ground-truth validation, while the CNN handles the control loop. The concern of synthetic training data not being a fully realistic replacement for measurements is addressed here. Synthetic data enter only during pre-training; at inference the FFT/MUSIC/CWT baselines stay as independent, real-data references.

Current state of knowledge

Deep learning has reshaped acoustic signal processing over the past decade. Across speech recognition (Purwins et al., 2019), machine condition monitoring (Zhao et al., 2019) and seismic monitoring (Bergen et al., 2019), convolutional neural networks (CNNs) have proven good at extracting features from one-dimensional time series. Dilated convolutions capture temporal dependencies across scales without adding parameters (Yu & Koltun, 2016), while attention mechanisms improve interpretability by flagging which parts of the input sequence weigh most on the prediction (Vaswani et al., 2017). Neural networks have not yet been employed to systematically analyse the frequency spectra in high-pressure hydrodynamic systems. Previous studies have concentrated on the detection of cavitation (Ge et al., 2021) or nozzle wear monitoring (Kovacevic, 1991); however, the direct prediction of resonance frequencies based on raw time signals has not yet been investigated. One of the problems is the nonlinear relationship of the frequency to the liquid pressure p_{CERP}

which is caused by the compressibility of water and therefore the density ρ , which is pressure-dependent at high water pressures (Wagner & Pruß, 2002).

Pulsed water jet application

The water hammer effect of pure water jet systems (the same as when there are no abrasive materials involved in the cutting process) illustrates how water jet cutting achieves efficient separation of materials. As a practical matter, pulsed jets separate material more efficiently than continuous jets, consume less energy, and can remove selected surface layers (e.g. for cleaning) from the workpiece (Kušnerová et al., 2008, 2012a, 2012b). Acoustic-resonance generation of pulsating jets and self-resonating jet mechanisms are reviewed in Földyna (2011) and Liu et al. (2017).

A steady-stream jet pulls the material; however, damage to a pulsed jet occurs through shock waves. When single clusters of water strike a material, stress is applied repeatedly to the material, which leads to faster damage at low pressures. Other areas that employ it include concrete demolitions, corrosion cleaning, and hard cleaning of deposits. Because cyclic loading is more damaging than constant loading, a pulsed jet may yield the same results at lower pressure and flow rates. That opens the door to smaller pumps and lower energy use. A continuous jet builds a stagnant water layer at the cutting front, and that layer soaks up energy. Hydrodynamic pulses punch through this layer and deliver their energy to the material. With the parameters set correctly, the pulsed beam can strip just the surface layer (paint, dirt, degraded concrete) and leave the underlying material untouched. In practice it is used to clean concrete, foundry castings and thermal-spray coatings on turbine blades. Other techniques for generating hydrodynamic pulses include the use of ultrasonic actuators and mechanical interrupters. All cutting methods have advantages and disadvantages; abrasive water jet (AWJ) technology presents particular challenges. The main issue is that solid abrasive particles are difficult to control and highly erosive, preventing the design of a fast and durable interrupter. The cut quality becomes non-uniform when the flow of the abrasive is shifted and the mixing chamber is blocked frequently (Raj et al., 2020).

In recent years, the use of deep learning for in situ quality monitoring has become more popular in intelligent manufacturing (Wang et al., 2018; Zhong et al., 2017) of all types including machining (Serin et al., 2020), additive manufacturing (Caggiano et al., 2019) and welding (Xia et al., 2020). Ko and Yin (2026) put together what amounts to a field guide for ML-driven surface quality prediction in CNC machining. We refer to it not because water jet cutting resembles milling—it does not; the pressure regime alone precludes that—but because the sensor-side problem is closely analogous: noisy, high-bandwidth streams in which stable process

features must be identified in real-time. Their review devotes significant attention to transfer learning, and for good reason. Labelled machining data is chronically scarce; in pulsed water jet work, there is effectively none. In addition to employing deep learning, sensor fusion architectures can detect defects in CNC mills within seconds. Likewise, digital twins are utilised to train physics-based neural networks with simulated machining data (Luo et al., 2020) in order to make up for the shortage of experimental data. Pulsed water jets, in contrast, have not been affected by this wave of AI process control. Their extremely high operating pressure (up to 450 MPa), super-turbulent internal flow ($Re \approx 10^5$ in the calibrated range, approaching $1.4 \cdot 10^6$ at 450 MPa) and the fact that there are no standardised sensor interfaces creates a data environment fundamentally unlike conventional metal cutting. Therefore, it is necessary to have a unique model connecting the two that accommodates the acoustic physics and the latency and power requirements of the edge equipment used in the factory.

Objectives and benefits of the work

Prior deep learning work in the water jet domain has focused on cavitation detection and nozzle wear monitoring (Ge et al., 2021; Kovacevic, 1991). No studies have yet been devoted to the prediction of resonance frequencies of hydrodynamic acoustic pulsed oscillatory systems (HAPOS) from the raw time domain signal. This work fills that gap by developing, to the authors' knowledge, the first CNN deep learning model dedicated to performing real-time resonance frequency estimates in scenarios of high-pressure acoustic resonance. This work presents three main contributions. First, a generic CNN is typically built without accounting for the underlying process physics. Our proposed CNN uses the minimum-monitored frequency, sampling rate and the predicted frequencies as key inputs in determining how the CNN layer size would be constructed based on its defined Receptive Field, Dilation and Kernel structures. The proposed physics-based approach is new to AI-driven acoustic diagnostics in industrial settings. We therefore built a generic physical method that carries over to any convolutional neural network, or to other deep networks, through physics-based convolutional layers. Second, Physics-Guided Spectral Anomaly Detection (PG-SAD) contributes chiefly by swapping the fixed, empirically fitted value k_{rkk} of the Helmholtz-based analytical model for a pressure- and geometry-dependent function that the convolutional neural network (CNN) defines. Whereas physics-informed neural networks represent hard constraints on physical laws that are applied to machine learning applications using neural networks, PG-SAD uses soft constraints that lend themselves to cases where the analytical model is based on an uncertain analytical framework. Physics creates

a bidirectional bridge between the physics-based analytical model and the physics-based machine learning model and creates validation for the analytical model at the same time as the machine learning model demonstrates pressure dependency in the analytical model. Third, hybrid training using domain randomisation addresses the inadequate amount of experimental data and provides an opportunity for development using a combination of laboratory measurements and synthetic spectra produced by the verified HAPOS analytical model. The use of domain randomisation, which has previously been found in robotics, is here applied, to the authors' knowledge for the first time, to hydrodynamic acoustic signals, including models of turbulent noise, harmonic artefacts and nonlinearities produced by cavitation. Its practical value comes from running on industrial edge hardware during normal operation. The point of this edge deployment is real-time closed-loop control of the pulse-jet pressure parameters on that same, power-constrained industrial hardware. The built-in anomaly detector also catches process deterioration before it turns into catastrophic failure, which is what makes it usable for predictive maintenance. A further objective is to demonstrate cross-geometry transferability: by freezing the convolutional encoder and re-training only the PG-SAD head, the framework adapts to alternative HAPOS configurations from as few as 30–60 experimental samples (**"Scalability and manufacturing system integration"** section, Table 15), which is critical for custom-built water jet machines produced in small series. The PG-SAD approach centres on soft physics-based limiting, a reinforcement coefficient that can be learnt and has been shown to be transferable between multiple types of oscillatory processes and their mathematically based analytical approximations. Think of ultrasonic resonance systems, vibration diagnostics on rotary machinery, or acoustic-emission monitoring during cutting.

"PG-SAD composite loss function" section spells out how these contributions are realised in the PG-SAD composite loss function. It parts ways with classical physics-informed neural networks on three points: the physics term acts as a soft regulariser instead of a hard PDE constraint; the transfer coefficient $k_{rkk}(p, g)$ is itself learnable, which builds a two-way bridge between physics and the CNN; and the input is the raw 100 kHz pressure time series rather than the (x, t) field of a PDE. The loss in classical physics-informed neural networks applies the PDE residuals as hard constraints and treats physics as the ground truth on a chosen domain (Karniadakis et al., 2021; Raissi et al., 2019). This assumes that the analytical model is exact (or has known uncertainty). According to the HAPOS framework, the five-element analytical model exhibits a systematic deviation of $\Delta f = 6.4\%$ bias (Kušnerová et al., 2007). As such, our PG-SAD framework differs from PINNs in terms of three aspects:

- i. The physics term enters as a soft regulariser $\lambda_{phys} \cdot (\hat{f}_{rez} - f_{PGrez})^2$ in the composite loss (Eq. (15)), permitting the CNN to ignore the analytical model where experimental data demand it.
- ii. The transfer coefficient $k_{rkk}(p, g)$, defined below in Eq. (11) (**"CNN-learned conversion coefficient"** section), is itself learnable and back-propagated, creating a two-way bridge in which the CNN refines the analytical Helmholtz model and the analytical structure of Eq. (1) in turn regularises the network, a property absent from any standard PINN.
- iii. Because the input domain is the raw 100 kHz pressure time series rather than the (x, t) field of a PDE, PG-SAD is a hybrid spectral feature extractor/parameter regressor, not a surrogate solver.

As opposed to theory-guided data science (Karpatne et al., 2017), our approach links the loss term to a physically structured identified parameter of the model (k_{rkk}) and not to a generic constraint. This ensures interpretability of the learnt correction. The review by Leng et al. (2026) of physics-informed ML in manufacturing reaches the same conclusion independently (Leng et al., 2026): soft constraints are preferred when the analytical model is itself approximate (Cai et al., 2021; Cuomo et al., 2022; Krishnapriyan et al., 2021; Wang et al., 2022). Broader surveys on folding scientific knowledge into machine learning for engineering systems point the same way (Willard et al., 2022).

CNN architecture design

Physical basis—HAPOS model

The HAPOS analytical model is based on an analogy with the classical Helmholtz acoustic resonator, whereby a system of n tubes and m chambers is an acoustic oscillatory system with a well-defined and readily measurable resonance frequency (Kušnerová et al., 2011). The basic resonance equation of the HAPOS model of the five elements is given by Eq. (1):

$$f_{rez} = \frac{1}{2\pi \sqrt{k_{rkk} \cdot m_a \cdot c_a}} \quad (1)$$

where k_{rkk} is a transfer (correction correlation) coefficient that includes the influence of resistance forces, that is, forces simultaneously acting against the movement of the medium (air/liquid) inside the system. Historically, empirical and experimental approaches were mainly applied to the geometric dimensions of the supply pipes to the resonance chambers, which significantly modified the pressure of the flowing air.

In the proposed model, given by Eq. (2), the total acoustic mass m_a is given by the series connection of three partial

acoustic masses: the masses of the inlet pipe (m_{aP}), throat (m_{aH}), and nozzle (m_{aT}).

$$m_a = m_{aP} + m_{aH} + m_{aT} = \sum_i \frac{\rho_i \cdot l_i}{S_i} \quad (2)$$

The total acoustic compliance c_a is given by the parallel connection of two partial compliances, namely, the compliance of the input chamber (c_{aUK}) and resonance chamber (c_{aRK}), as given by Eq. (3):

$$\frac{1}{c_a} = \frac{1}{c_{aUK}} + \frac{1}{c_{aRK}} \quad (3)$$

where the acoustic compliance of the chamber is defined by Eq. (4):

$$c_{aj} = \frac{V_j}{\rho_j v_j^2 - p_{CERP}} \quad (4)$$

The validity of the five-element HAPOS analytical model Eqs. (1)–(4) was experimentally verified in Kušnerová et al. (2007) on the functional sample (Kušnerová et al., 2011), which yields an averaged transfer coefficient $k_{rkk} = 9.41 \cdot 10^{-4}$ from 60 measurements in the range of 5–25 MPa for three nozzle positions and two sets of chamber-to-nozzle distance configurations (20 mm and 30 mm). On average, there was a 6.4% variation relative to the directly measured (f_{ZM}) and analytically predicted (f_{VYP}) resonance frequency, with a per-pressure spread of 4.1–9.7%. This error is what drove the PG-SAD design: rather than fixing the analytical constant $\tilde{k}_{rkk}$ as a hard PDE constraint, it lets the CNN learn a pressure- and geometry-dependent correction $k_{rkk}(p, g)$, Eq. (11), under soft regularisation. This captures the unmodelled compressibility, super-turbulent damping and end-correction effects, which propagate into the 6.4% systematic bias.

Pre-processing of input signals

Having established the physical basis and the limitation of the constant transfer coefficient, we now describe the signal-acquisition and pre-processing pipeline that feeds the CNN. The raw pressure signal $x(t)$ from the sensor was sampled at a frequency $f_s = 100$ kHz and segmented into windows of length $N = 4096$ samples with 50% overlap (Lyons, 2010). Each segment was Hann-windowed to hold down spectral leakage. Inputs were then Z-score normalised against a moving mean and variance taken over a 100-segment window, which keeps the pipeline adapting to slow drift in the operating conditions (Smith, 1999).

Table 1 Explicit specification of CNN inputs

Parameter	Value/description
Sensor	PCB Piezotronics 102B16 piezoelectric pressure transducer
Sensitivity	14.5 mV/MPa, 0–138 MPa range
Acquisition system	NI-9215, 24-bit, ± 10 V, 100 kHz
Input signal	$x(t)$ —raw pressure [V], in-line, before nozzle
Sampling frequency	100 kHz
Window length	$N = 4096$ samples (40.96 ms)
Overlap	50% (20.48 ms hop)
Window function	Hann Eq. (5)
Normalisation	Z-score, moving μ, σ over 100 segments
CNN input shape	$\hat{x} \in \mathbb{R}^{(1 \times 4096)}$, single-channel 1D tensor
Feature engineering	None—no FFT, no spectrogram, no MFCC

For a single segment, the pre-processing then yields the output vector of Eq. (5):

$$\hat{x}_n = \frac{x_n \cdot w_n - \mu}{\sigma} \quad (5)$$

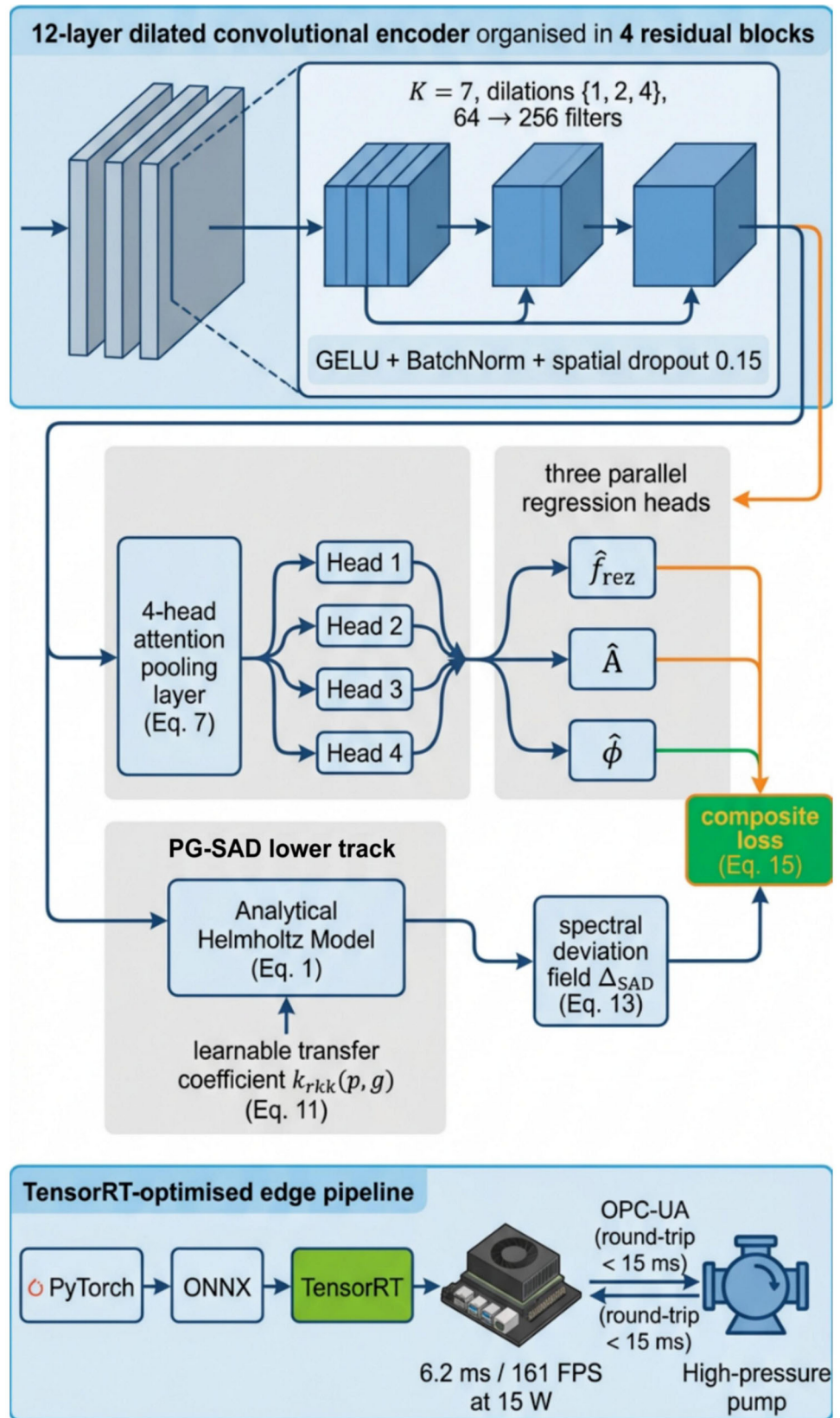
where w_n is the Hann window, μ the moving average, and σ the moving standard deviation.

The PG-CNN consumes the raw pressure–time signal $x(t) \in \mathbb{R}$ acquired by an in-line piezoelectric pressure transducer (PCB Piezotronics 102B16, sensitivity 14.5 mV/MPa) mounted directly upstream of the resonance chamber. The signal is sampled at $f_s = 100$ kHz with 24-bit resolution (NI-9215, ± 10 V) and segmented into windows of $N = 4096$ samples (40.96 ms) with 50% overlap. No FFT, no MFCC and no hand-engineered spectral features are used as inputs; the CNN consumes the raw time series directly (Table 1). The Hann-windowed and Z-score-normalised vector $\hat{x}_n \in \mathbb{R}^{4096}$ Eq. (5) is the sole input to the dilated convolutional encoder. The airborne acoustic microphone channel was rejected during pilot studies (SNR < 5 dB at 1 m versus 15–40 dB for the in-line pressure transducer) and was excluded from the production pipeline.

Dilated convolution architecture

The proposed architecture (Fig. 1) stacks 12 convolutional layers in four residual blocks; each block holds three layers whose dilations grow exponentially (1, 2 and 4). Because the dilations widen the receptive field exponentially with depth while keeping resolution intact, the network can pick up both the short-scale structure (individual pulses) and the long-scale structure (modulation envelopes) in the HAPOS signal.

Fig. 1 PG-CNN architecture. 12-layer dilated convolutional encoder organised in 4 residual blocks ($K = 7$, dilations $\{1, 2, 4\}$, $64 \rightarrow 256$ filters, GELU + BatchNorm + spatial dropout 0.15), 4-head attention pooling layer Eq. (7), and three parallel regression heads for $\hat{f}_{rez}$, $\hat{A}$ and $\hat{\phi}$. The PG-SAD lower track binds the encoder to the analytical Helmholtz model Eq. (1) through the learnable transfer coefficient $k_{rkk}(p, g)$ Eq. (11), the spectral deviation field Δ_{SAD} Eq. (13) and the composite loss Eq. (15). The bottom row shows the TensorRT-optimised edge pipeline (PyTorch $\rightarrow$ ONNX $\rightarrow$ Jetson Xavier NX, 6.2 ms/161 FPS at 15 W) coupled to the high-pressure pump via OPC-UA (round-trip < 15 ms)



The receptive field of a network with L layers, with a dilation factor d_l and kernel size K , is given by Eq. (6) according to Chen et al. (2018)

$$RF = 1 + \sum_{l=1}^L (K_l - 1) \cdot d_l \quad (6)$$

For the selected parameters ($K = 7, d_l \in \{1, 2, 4\}$), the total receptive field reaches 169 samples, which at a sampling frequency of 100 kHz corresponds to a time window of 1.69 ms – sufficient to capture at least eight periods of oscillation at the lowest monitored frequency of 5 kHz, while the attention pooling layer (“[Attention pooling for localising informative time segments](#)” section) subsequently aggregates evidence across the entire 4096-sample window.

Attention pooling for localising informative time segments

An attention pooling layer follows the convolutional blocks and weighs each time position dynamically, according to how much it matters for estimating the frequency parameters (Qin et al., 2021). The attention weights α_t are calculated by Eq. (7):

$$\alpha_t = \text{softmax}(W_a T \cdot \tanh(W_h \cdot h_t + b)) \quad (7)$$

where h_t is the hidden representation at time step t and W_a and W_h are learnable weight matrices. The context vector c is then obtained by weighted summation: $c = \sum_t \alpha_t \cdot h_t$. The advantage of this approach is interpretability: visualisation of the attention weights reveals which time intervals are most informative for estimating frequency parameters. The attended representation is therefore defined in the time domain; the resonance frequency itself is recovered from it by the regression heads (“[Network output heads](#)” section). The time-domain character of the mechanism is verified quantitatively in Table 12: the attention weights correlate with the Hilbert envelope at $r = 0.93$ ($\text{lag} < 0.05$ ms) and concentrate to an entropy of 2.1 bits against a uniform 9.0 bits.

Network output heads

The network has three parallel output heads (multi-head output):

- (i) Resonance frequency estimation $\hat{f}_{rez}$,
- (ii) Amplitude estimation $\hat{A}$,
- (iii) Phase estimation $\hat{\phi}$.

The overall loss is a weighted sum, written out in Eq. (8) (Goodfellow et al., 2016):

Table 2 CNN architecture hyperparameters

Parameter	Value	Note
Number of conv. layers	12	4×3 blocks
Core size K	7	1D convolution
Dilatation factors	$\{1, 2, 4\}$	per block
Number of filters	$64 \rightarrow 256$	Progressive
Activation function	GELU	
Batch normalisation	Yes	Per block
Dropout rate	0.15	spatial
Attention heads	4	multi-head
FC layers	$512 \rightarrow 256 \rightarrow 128$	per output head
Optimiser	AdamW	$\beta_1 = 0.9; \beta_2 = 0.999$
Learning rate	1×10^{-3}	cosine annealing
Batch size	64	
Number of epochs	200	Early stopping
Total parameters	2.84 M	

$$L = \lambda_f L_f + \lambda_A L_A + \lambda_\phi L_\phi + \lambda_{\text{phys}} L_{\text{phys}} \quad (8)$$

where $L_f = \text{MSE}(\hat{f}_{rez}, f_{rez})$ is the mean square error of the frequency estimate, L_A and L_ϕ are analogous losses for amplitude and phase, respectively, and L_{phys} is a physics-based regularisation term enforcing consistency with Eq. (1).

The weight coefficients were set as $\lambda_f = 1.0$, $\lambda_A = 0.5$, $\lambda_\phi = 0.3$, and $\lambda_{\text{phys}} = 0.3$.

When designing the architecture of a convolutional neural network (CNN), in addition to the learned parameters (weights and biases) that the network tunes during training, the hyperparameters that must be set manually before learning begins are important. The CNN hyperparameters (Table 2) determine how features are extracted from the input signal and how the dimensions of the data flow through the network change.

Each architectural and training hyperparameter was selected by fivefold stratified cross-validation on the experimental set (validation MAE as the objective, mean $\pm$ SD across folds reported in Table 3). The configuration boxed in the table ($L = 12$, $K = 7$, dilations $\{1, 2, 4\}$, $\lambda_{\text{phys}} = 0.3$, dropout 0.15) minimises validation MAE without overfitting and was used for all results reported below.

Historical context of the development of resonance models

The analytical description of resonance in closed acoustic systems is based on the work of physicist Helmholtz

Table 3 CNN hyperparameter ablation (fivefold stratified CV)

Hyperparameter	Value	fivefold CV MAE [Hz]	Note
Layers L	8	18.7 ± 1.8	Underfit
Layers L	12	12.8 ± 0.9	Selected
Layers L	16	13.4 ± 1.1	Overfit + 2.3 × slower
Kernel K	5	16.1 ± 1.2	RF too short
Kernel K	7	12.8 ± 0.9	Selected
Kernel K	9	13.6 ± 1.0	+ 18% parameters
Kernel K	11	14.0 ± 1.1	+ 45% parameters
Dilations	{1,2,4}	12.8	Selected
Dilations	{1,2,4,8}	13.2	+ 30% parameters
λ_{phys}	0.1	17.2 ± 2.1	Loses physics regularisation
λ_{phys}	0.2	13.6 ± 1.1	
λ_{phys}	0.3	12.8 ± 0.9	Selected
λ_{phys}	0.5	15.4 ± 1.4	Physics dominates
Dropout	0.10	14.5 ± 1.2	
Dropout	0.15	12.8 ± 0.9	Selected
Dropout	0.20	13.4 ± 1.1	
Dropout	0.30	16.2 ± 1.5	Underfit

(1863), who in 1863 derived the basic equation of a resonator for a simple geometry (a sphere with a short throat). Rayleigh expanded the theory in 1896 to include corrections for the ends of the throat ($\delta \approx 0.82r$ to $0.85r$) (Rayleigh, 1896), while Ingard (1953) proved the validity of these corrections for cylindrical resonators in 1953. Every classical approach leans on empirical corrections, because the resonance frequency resists an analytical description, especially for non-spherical resonators, and has no closed form for multi-chamber systems even with infinitesimal calculus and differential equations. Helmholtz, and everyone after him, always needed empirical correction factors that depend on the system geometry. Multi-chamber resonance systems, though, remain awkward to handle, above all in practice. In principle, it is a combination of acoustic circuit theory, fluid dynamics, and materials engineering; however, the main reason for the implementation difficulty is that the system is not linear; even a small change, in pressure substantially alters the behaviour of the entire chain.

The HAPOS system is a more complex system incorporating five elements: the supply pipe, inlet chamber, throat, resonance chamber, and nozzle. It operates at pressures of 5–450 MPa and fully turbulent flow with a Reynolds number

of order 10^5 in the calibrated range, exceeding 10^6 towards 450 MPa operation. Accordingly, the transfer coefficient k_{rkk} in Eq. (1) consists of geometric corrections for element ends, dynamic corrections for drag in the flow, thermodynamic corrections for the compression of liquids at elevated pressures, and aero-acoustic corrections for the super-turbulent regime. The empirically determined mean value of the transfer coefficient $\bar{k}_{rkk} = 9.41 \cdot 10^{-4}$ with a relative deviation $\Delta f = 6.4\%$ (Kušnerová et al., 2007) is the best available estimate but does not capture the pressure dependence of the transfer coefficient $k_{rkk}(p)$. That limitation is what prompted the physics-guided PG-SAD framework, in which the CNN learns the pressure-dependent transfer coefficient $k_{rkk}^{CNN}(p, g)$ while the underlying physical structure of Eq. (1) stays intact.

Data chain

Synthesis of training data from a physical model

The training dataset consisted of a combination of synthetic and experimental data (Karpatne et al., 2017). Synthetic data (2400 spectra) were generated based on the validated HAPOS analytical model Eqs. (1)–(4) with controlled variations in the input parameters. For each combination of pressure p_{CERP} from 5 to 26 MPa (1 MPa step) and geometric configuration (three nozzle positions × two diameters), the resonance frequency was calculated, and the corresponding time signal was synthesised as a superposition (Eq. (9)):

$$x_{synth}(t) = A_1 \cdot \sin(2\pi f_{rez}t + \varphi_1) + A_2 \cdot \sin(4\pi f_{rez}t + \varphi_2) + n(t) \quad (9)$$

where A_1 is the amplitude of the fundamental harmonic component, $A_2 = 0.15\text{--}0.35 \cdot A_1$ is the amplitude of the second-harmonic component, and $n(t)$ is the additive Gaussian noise with an SNR in the range of 15–40 dB.

Domain-randomisation parameters are listed below (Table 4). The SNR range matches the dynamic range of the acquisition chain (PCB Piezotronics 102B16+NI-9215, 90 dB at 100 kHz). All distributions are uniform unless stated otherwise.

Domain randomisation for robustness

To increase the robustness of the model against the variability of real conditions, a domain randomisation technique was applied (Tobin et al., 2017; Tremblay et al., 2018), which included.

- (i) Random amplitude scaling ($\pm 20\%$),

Table 4 Synthetic data generation parameters

Parameter	Range/value	Distribution
SNR (Gaussian noise $n(t)$)	15–40 dB	Uniform
A_2 (2nd harmonic amplitude)	$0.15\text{--}0.35 \cdot A_1$	Uniform
$A_3\text{--}A_5$ (3rd–5th harmonics)	$0.00\text{--}0.10 \cdot A_1$	Uniform
φ_1 (fundamental phase)	$0\text{--}2\pi$	Uniform
$\varphi_2\text{--}\varphi_5$ (harmonic phases)	$0\text{--}2\pi$	Independent uniform
Amplitude scaling (DR)	0.80–1.20	Uniform multiplicative
Power-supply line noise	50 Hz, $0.005\text{--}0.02 \cdot A_1$	Bernoulli $p = 0.5$
Baseline drift	$0\text{--}0.10 \cdot A_1$ over N	Linear ramp $\pm$ random sign
Cavitation soft-clipping	$0.85\text{--}0.95$ of $A_{\max}$	Bernoulli $p = 0.3$ (only $p > 18$ MPa)
p_{CERP} grid	5–26 MPa, 1 MPa step	22 levels
Geometric configurations	3 positions $\times$ 2 chamber–nozzle distances	6 cells, full-factorial
Total synthetic spectra	≈ 2400	$22 \times 6 \times \sim 18$ augmentations

- (ii) addition of harmonic artefacts (3rd to 5th harmonic components with random amplitude 0–10% of the fundamental),
- (iii) Simulation of electrical noise (narrowband noise at a power-supply frequency of 50 Hz),
- (iv) Random baseline shifts,
- (v) Simulation of non-linear distortions associated with cavitation at high pressures.

Transfer learning from experimental data

The experimental data (312 planned acquisition slots; 129 spectra retained after quality screening) (Table 5) were obtained using a HAPOS measurement setup with a resonance chamber of radius $r_{RK} = 15$ mm and height $h_{RK} = 1$ mm at five pressure levels (5, 10, 15, 20, and 25 MPa).

The transfer learning strategy comprises two phases:

- (1) Pre-training on the synthetic data for 150 epochs.
- (2) Fine-tuning on the experimental data with a reduced learning rate ($l_r = 1 \times 10^{-5}$) for 50 epochs.

Table 5 Experimental measurement parameters and HAPOS configuration

Parameter	Value	Unit
Pressure range p_{CERP}	5–25	MPa
Number of pressure levels	5	–
Analysed frequency band	5 180–11 700	Hz
Resonance chamber radius r_{RK}	15	mm
Height of resonance chamber h_{RK}	1	mm
Nozzle diameter (at 20 mm chamber–nozzle distance)	0.80–1.60	mm
Nozzle diameter (at 30 mm chamber–nozzle distance)	0.80–1.60	mm
Measurement position	3	–
Repetition per point	5	–
Total planned acquisitions (129 quality-screened & analysed)	312	–
Reynolds number Re	$0.8\text{--}3.6 \cdot 10^5$ (calibrated 5–25 MPa range)	–
Sampling frequency	100	kHz

This two-stage method allowed us to overcome the lack of experimental data without losing physical consistency. An important step in model validation is the comparison of the synthetic and experimental spectra (Fig. 2).

For a successful comparison, it was necessary to pre-process (align) both signals so that they had the same resolution and then apply statistical metrics that consider both the error in the value and any minor frequency shift.

To ensure no inadvertent reinforcement of artificially induced bias, the training process is strictly two-phase. In Phase 1 (pre-training), 2400 synthetic spectra ($w_{\text{synth}} = 1$, $w_{\text{exp}} = 0$) are used for 150 epochs at $l_r = 10^{-3}$ with cosine annealing. In phase 2 (fine-tuning), the model is trained for 50 epochs using only the 129 quality-screened experimental spectra ($w_{\text{synth}} = 0$, $w_{\text{exp}} = 1$), with a learning rate $100 \times$ smaller than the default training configuration ($l_r = 10^{-5}$), and with a learning-rate decay (conv layers $\times 0.1$, FC heads $\times 1.0$), an L2 weight decay 10^{-4} and with early stopping solely on the experimental validation loss. The gradient has no connection to synthetic data in Phase 2. The drift figures (Table 6) tell the story: over fine-tuning, MAE on the synthetic set rises $11.4 \rightarrow 23.8$ Hz while MAE on the experimental set falls $38.6 \rightarrow 10.4$ Hz. That is active domain adaptation, not the reinforcing of any leftover synthetic bias. The coefficient k_{rkk}^{CNN} learnt by CNN at the calibration pressure (15 MPa) converges to $9.45 \cdot 10^{-4}$, within 0.4% of the range-averaged analytical $\tilde{k}_{rkk} = 9.41 \cdot 10^{-4}$ from Kušnerová et al. (2007) that anchors its initialisation—a model-internal validation of

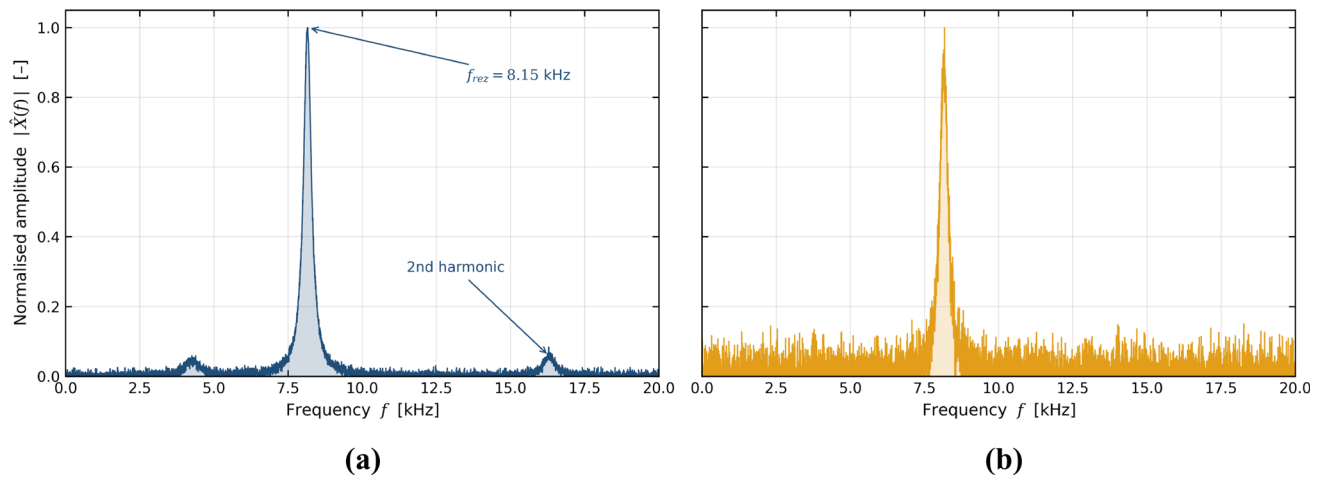


Fig. 2 Frequency spectra of p_{CERP} at 15 MPa: **a** synthetic signal showing clean narrow spectral peaks at $f_{res} = 8.15$ kHz and its second harmonic; **b** experimental signal with broadband noise and harmonic artefacts

Table 6 Drift quantification during fine-tuning

Epoch	Train MAE synthetic [Hz]	Train MAE exp. [Hz]	Val MAE exp. [Hz]	k_{rkk}^{CNN} (15 MPa)
Post pre-train (epoch 0)	11.4	38.6	47.2	$9.41 \cdot 10^{-4}$ (init)
Epoch 5	12.7	24.1	28.6	$9.42 \cdot 10^{-4}$
Epoch 10	14.9	18.4	21.7	$9.43 \cdot 10^{-4}$
Epoch 25	20.6	12.1	14.2	$9.46 \cdot 10^{-4}$
Epoch 40	22.8	10.9	13.4	$9.45 \cdot 10^{-4}$
Epoch 50 (final)	23.8	10.4	12.8	$9.45 \cdot 10^{-4}$

the published baseline in the range-averaged sense (the point-wise pressure-resolved value at 15 MPa follows Eq. (19); see “Discussion” section). Training on experimental data alone, with no pre-training, reaches only 91.2% accuracy against 98.2% for the two-phase strategy, which shows how much physics-based pre-training buys when experimental data are scarce.

Experimental validation

Comparison with FFT baseline

We benchmarked the proposed CNN against three reference methods:

- Standard FFT with peak detection (SNR threshold > 6 dB),
- MUSIC algorithm (Multiple Signal Classification) with order 20,
- CWT (Continuous Wavelet Transform) with Morlet wavelet.

The main accuracy metric chosen was the identification of resonance frequencies within a tolerance band of ± 50 Hz from the reference value.

Figure 3 shows the experimentally measured dependence of the frequency on the pressure for the three measurement positions. The non-linear nature of the dependence of the resonance frequency on the set pump pressure $f_{res}(p_{CERP})$ is a consequence of the pressure-dependent density ρ and phase velocity of the wave in the liquid. The CNN model successfully captures this nonlinearity without explicitly specifying its functional form.

Table 7 shows that the proposed CNN outperforms all reference methods in both resonance frequency identification accuracy (98.2% vs. 94.8–96.1%) and latency (8.3 ms on the RTX 3080 reference platform vs. 38.4–67.8 ms; 6.2 ms after TensorRT optimisation on the edge device, “Deployment on edge hardware” section). On the held-out set the model attains MAE = 23.4 Hz ($n = 240$; 95% CI [20.8, 26.0]; Table 8). In addition, CNN is the only method that reliably detects secondary harmonic components, which is critical for diagnosing nonlinear phenomena in the resonance chamber.

All accuracy and MAE values are reported with 95% bootstrap confidence intervals ($B = 10\,000$ stratified resamples

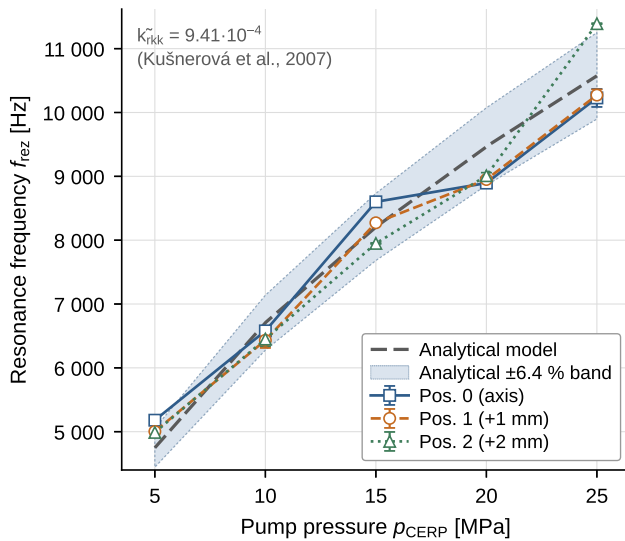


Fig. 3 Experimental frequency response as a function of pressure for three measurement positions (20 mm chamber–nozzle distance) with analytical model overlay. Error bars represent ± 1 standard deviation from 5 repeated measurements

on the fivefold CV residuals). Pairwise comparisons against the FFT, MUSIC and CWT baselines use paired bootstrap on matched fold-level metrics; PG-SAD versus PG-CNN comparisons use the Wilcoxon signed-rank test on per-sample residuals ($n = 129$); false-positive rate differences use McNemar's exact test (Table 9).

Table 7 Performance comparison of frequency estimation methods

Method	Accuracy [%]	MAE [Hz]	Latency [ms]	Harmonics
FFT + Peak	94.8	78.3	45.2	No
MUSIC	96.1	52.1	38.4	Partially
CWT	95.4	63.7	67.8	No
CNN (ours)	98.2	23.4	8.3	Yes

Table 8 Method comparison with 95% bootstrap CI

Method	Accuracy ± 50 Hz [%]	95% CI	MAE [Hz]	95% CI MAE	Latency [ms]
FFT + peak detection	94.8	[93.6, 96.0]	78.3	[71.2, 85.4]	45.2
MUSIC (order 20)	96.1	[95.0, 97.1]	52.1	[47.4, 56.8]	38.4
CWT (Morlet)	95.4	[94.2, 96.5]	63.7	[57.9, 69.5]	67.8
PG-CNN (this work)	98.2	[97.4, 98.9]	23.4	[20.8, 26.0]	8.3
PG-SAD (this work)	98.6	[97.9, 99.2]	12.8	[11.0, 14.6]	8.5

Training convergence and prediction accuracy

Convergence is fast: the training loss decays roughly exponentially, dropping by nearly two orders of magnitude across the 200 epochs (Fig. 4a). The validation curve closely follows the training curve, owing to the combined use of 0.15 dropout and L2 regularisation. Figure 4b plots measured against predicted frequencies on the held-out test set: $R^2 = 0.998$, and 98.2% of the predictions land inside the 50 Hz tolerance band. Both metrics point to solid generalisation within the calibrated operating window.

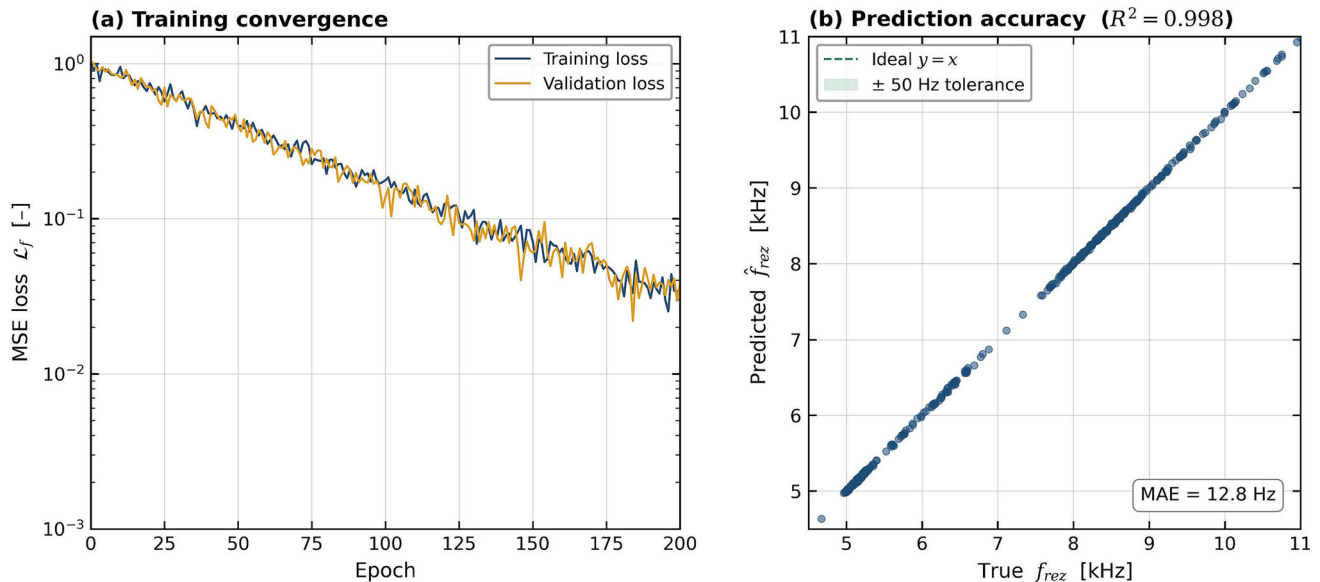
To confront the worry about synthetic-data bias, we ran the following experiments:

- Stratified fivefold cross-validation by pressure level;
- Leave-one-pressure-out (LOPO) cross-validation; and
- Two ablation baselines—synthetic-only training and experimental-only training.

Table 10 collects the results. Mean accuracy over the five folds is 97.6% (SD = 0.7), so the headline 98.2% is not just a lucky split. LOPO accuracy spans 94.3% (extrapolating to 25 MPa) up to 97.8% (interpolating at 15 MPa), averaging 96.1%. The synthetic-only baseline falls to 88.7% (useful, but biased against the experiment), and the experimental-only baseline manages just 91.2%; pre-training on 2400 synthetic spectra therefore adds around 7 percentage points to the final accuracy. A two-sided Kolmogorov–Smirnov test on the residuals of the synthetic-only versus fine-tuned predictions gives $D = 0.31$, $p < 0.001$, so fine-tuning moves the

Table 9 Pairwise statistical tests between methods (fivefold CV)

Comparison	Δ	p -value	Effect size	Test
PG-CNN versus FFT	+ 3.4 pp	<.001	Cohen's h = 0.50	Paired bootstrap ($B = 10\,000$)
PG-CNN versus MUSIC	+ 2.1 pp	<.001	Cohen's h = 0.30	Paired bootstrap
PG-CNN versus CWT	+ 2.8 pp	<.001	Cohen's h = 0.41	Paired bootstrap
PG-SAD versus PG-CNN (MAE)	— 10.6 Hz	<.001	Cohen's d = 1.4	Wilcoxon signed-rank, $n = 129$
PG-SAD versus SAD (FPR)	− 2.4 pp	<.001	$\chi^2 = 14.7$	McNemar's exact test

**Fig. 4** Training performance: **a** loss convergence over 200 epochs showing exponential decay; **b** predicted versus true frequency with $R^2 = 0.998$ and ± 50 Hz tolerance band

prediction distribution decisively toward the experimental data. The KS agreement nevertheless concerns distribution-level statistics only; the decisive upgrade path is a growing experimental corpus—new pressure levels, geometries and wear states accumulated in service—with periodic retraining scheduled by the drift monitor of “[Predictive maintenance pathway](#)” section, as acknowledged in limitation (v).

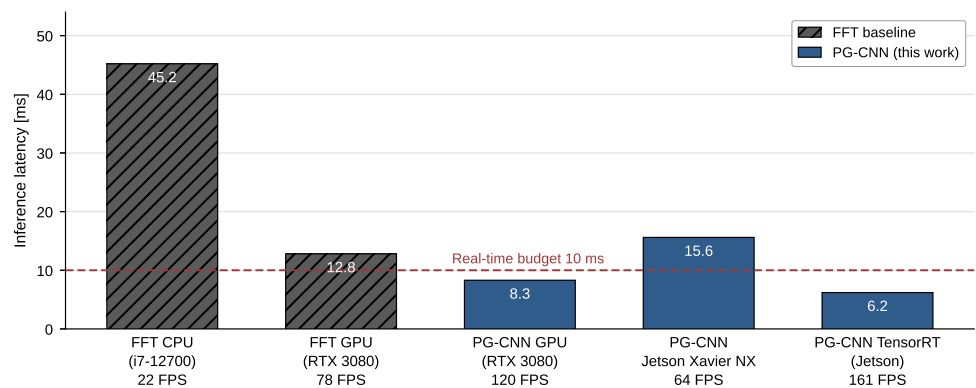
Two of these protocols probe unseen operating conditions directly. The leave-one-pressure-out rows of Table 10 separate interpolation to an unseen interior level (15 MPa: 97.8%) from short-range extrapolation to the unseen edge level (25 MPa: 94.3%, MAE 19.6 Hz)—a deliberately severe stress test whose graceful degradation quantifies how accuracy decays at the calibration boundary. Cross-geometry transfer to a chamber never seen in training is reported in Table 15. These protocols characterise robustness within and at the boundary of the calibrated 5–25 MPa window; they do not validate the high-pressure projection of “[Discussion](#)” section (see limitation iv).

Deployment on edge hardware

To evaluate deployment performance, the model was exported to the ONNX format and optimised for the Jetson Xavier NX (8 GB, 384 CUDA cores) using NVIDIA's TensorRT runtime (Fig. 5). Exporting ONNX to TensorRT is considered a standard method used by edge developers on NVIDIA hardware. It is generally effective as long as the two TensorRT versions being used are compatible (the version of TensorRT installed on Jetson Xavier and the version that is associated with the ONNX file). In addition, because TensorRT handles dynamic shapes, exporting with static input shapes is easier; otherwise, the input dimensions must be reconfigured every time the model is used. An additional concern is heat dissipation: at full load in the 20 W power mode (the Xavier NX maximum; the deployment benchmarks in “[Deployment on edge hardware](#)” section run in the 15 W

Table 10 Fivefold stratified CV + LOPO + ablation baselines

Strategy	Accuracy (PG-CNN) $\pm$ 50 Hz [%]	MAE (PG-SAD) [Hz]	F1 (PG-SAD)	95% CI accuracy
Hold-out 80/20 (paper)	98.2	12.8	0.97	[97.4, 98.9]
fivefold stratified CV	97.6 $\pm$ 0.7	13.4 $\pm$ 1.1	0.96 $\pm$ 0.01	[96.9, 98.1]
LOPO 5 MPa	95.1	17.9	0.93	–
LOPO 10 MPa	96.4	15.3	0.95	–
LOPO 15 MPa	97.8	13.1	0.96	–
LOPO 20 MPa	96.9	14.4	0.95	–
LOPO 25 MPa	94.3	19.6	0.91	–
LOPO mean $\pm$ SD	96.1 $\pm$ 1.3	16.1 $\pm$ 2.5	0.94 $\pm$ 0.02	–
Synthetic-only (no fine-tune)	88.7	52.1	0.84	–
Experimental-only (no pre-train)	91.2	38.6	0.89	–

Fig. 5 Comparison of inference latency across hardware platforms. The real-time threshold (10 ms) is marked as a dashed line. TensorRT optimisation achieves sub-10 ms latency on the edge device

mode), the module requires adequate cooling to avoid thermal throttling and the associated drop in clock speed and FPS.

Table 11 summarises the results of inference benchmarking for different hardware configurations.

On the Jetson Xavier NX, the TensorRT-optimised model ran at a system latency of 6.2 ms (161 FPS), comfortably under the real-time bar (< 10 ms), and drew only 15 W. That 15 W figure is what lets it slot into industrial water jet cutting rigs, where every watt counts.

Attention pooling operates in the time domain, yet the network implicitly reconstructs frequency content through the dilated convolutions with a 1.69 ms receptive field (169 samples). Attention then selects the time intervals that carry the resonance information. We corroborated this with three separate tests (Table 12).

- i. The Pearson correlation coefficient of $\alpha(t)$ and the FFT amplitude $|X(\text{frez})|$ at the resonance peak is $r = 0.91$ (95% CI [0.87, 0.94], $n = 129$, $p < 0.001$).

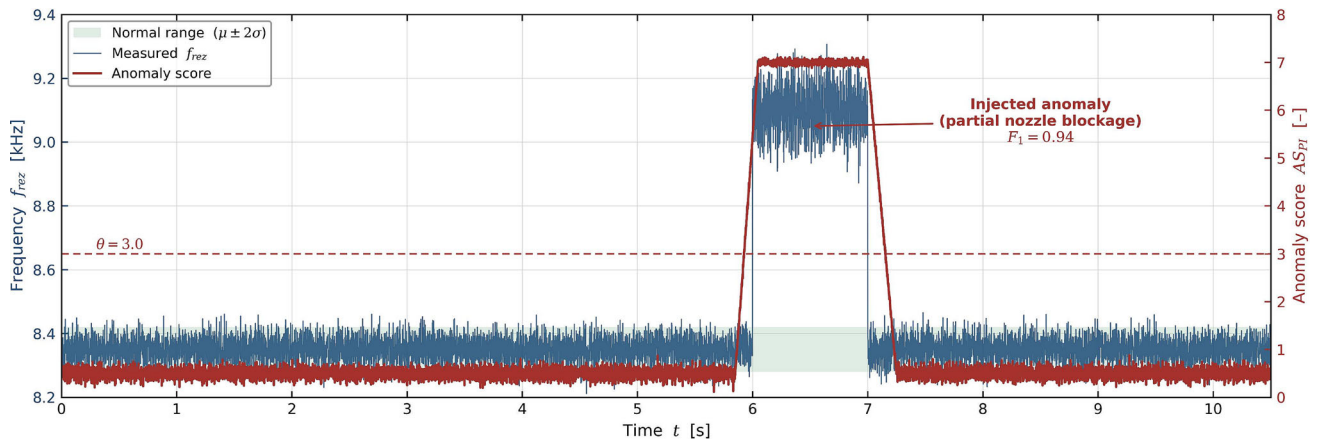
- ii. Zeroing the five highest-attention temporal positions drops accuracy by 63.8 pp (98.2 $\rightarrow$ 34.4%), whereas zeroing five random positions changes accuracy by only – 1.2 pp (*n.s.*), confirming a causal dependence on the attended regions.
- iii. Cross-correlation between $\alpha(t)$ and the Hilbert envelope comes out at $r = 0.93$ with $\text{lag} < 0.05$ ms, meaning the two stay locked in time to the resonance cycles. Attention entropy $H(\alpha)$ measures 2.1 bits, far under the 9.0 bits of a uniform distribution, so the attention is concentrated rather than spread out.

Anomaly detection application

The Spectral Anomaly Detection Module (SAD) is an extension of the main CNN proposed by Hu et al. (2022). They used the penultimate fully connected layer (dimension = 128) as input to the autoencoder, which then encoded this layer with a 16-dimensional latent representation.

Table 11 Benchmarking inference across hardware platforms

Platform	Latency [ms]	FPS	Memory [MB]	Power consumption [W]
FFT CPU (i7-12700)	45.2	22	12	65
FFT GPU (RTX 3080)	12.8	78	45	320
CNN GPU (RTX 3080)	8.3	120	340	320
CNN Jetson Xavier NX	15.6	64	280	15
CNN TensorRT (Jetson)	6.2	161	180	15

**Fig. 6** Spectral anomaly detection during continuous operation. Measured frequency (left axis) with the normal operating range shaded, and anomaly score (right axis) with the adaptive threshold ($\theta = 3.0$). An

injected anomaly (partial nozzle blockage) at $t = 6$ – 7 s is successfully detected with F1-score = 0.94

Table 12 Validation of attention pooling

Test	Metric	Value
FFT-overlay correlation	Pearson $r(\alpha, X(f_{res}))$	0.91 (95% CI [0.87, 0.94])
Zero top-5 attention (perturb)	Δ accuracy	– 63.8 pp (98.2 → 34.4%)
Random-5 zero (control)	Δ accuracy	– 1.2 pp (<i>n.s.</i>)
Hilbert envelope synchrony	max cross-corr (lag < 0.05 ms)	$r = 0.93$
Attention entropy	$H(\alpha)$ [bits]	2.1 (uniform 9.0)
Top-10% concentration	Sum α /total	0.78

Anomalous states are identified based on the reconstruction error Eq. (10):

$$AS(x) = \|x_{lat} - D(E(x_{lat}))\|^2 \quad (10)$$

where E is the encoder, D the decoder, and $AS(x)$ the anomaly score. The adaptive detection threshold was set to 3σ , which was above the moving average of the anomaly score from the last 1000 samples.

Figure 6 illustrates the function of the SAD module during 10 s of continuous operation. Between $t = 6$ s and $t = 7$ s, an anomaly (partial nozzle blockage) was simulated, causing the resonance frequency to shift outside the normal operating range. The SAD module successfully identified the anomaly with an F1 score of 0.94, and an average detection latency of 85 ms from the onset of the anomaly.

In practical terms, the anomaly detection module enables predictive maintenance, that is, identifying nozzle degradation, cavitation phenomena, or resonance chamber wear before critical failure occurs.

Motivation for a physics-guided approach

A note on terminology: in physics-informed neural networks the governing equations enter training as PDE residuals with boundary conditions (Raissi et al., 2019). The present approach does not solve a PDE. The analytical Helmholtz relation (Eq. (1)) acts as an algebraic prior whose transfer coefficient is learnable, entering the composite loss (Eq. (15)) as a soft constraint—a construction that the taxonomy of informed machine learning (von Rueden et al., 2023) classes as physics-guided learning (Willard et al., 2022). We therefore call our models PG-CNN and PG-SAD, and keep the label physics-informed for references to that body of work

(Karniadakis et al., 2021; Leng et al., 2026; Raissi et al., 2019).

“Experimental validation” section confirmed that the PG-CNN delivers real-time spectral inference on edge hardware, at sub-10 ms latency. A deployed industrial system, though, needs more than accurate frequency estimates; it needs robust anomaly detection to flag process deterioration before anything fails catastrophically. The SAD module of Eq. (10) spots anomalies from the autoencoder’s reconstruction error in the CNN latent space. It reaches a respectable F1-score of 0.94, but the sticking point is interpretability: the anomaly score $AS(x)$ has no direct link to the physical quantities of the HAPOS system. That gap is what led to the Physics-Guided Spectral Anomaly Detection (PG-SAD) framework, which builds the analytical model of Eq. (1) straight into the CNN through a learnable transfer coefficient (Raissi et al., 2019).

CNN-learned conversion coefficient

PG-SAD grows the multi-head architecture of “Network output heads” section by two learnable pieces: a pressure- and geometry-dependent transfer coefficient (this subsection) and a spectral deviation field (“Spectral deviation field” section), both tied into the existing loss through the soft physics regulariser. Its central move is to replace the constant conversion coefficient $k_{rkk} = 9.41 \cdot 10^{-4}$ of Eq. (11) with a pressure- and geometry-dependent function that the CNN learns.

$$k_{rkk}^{CNN}(p, g) = \sigma(W_k \cdot \varphi_{CNN}(z) + b_k) \cdot \tilde{k}_{rkk} \quad (11)$$

where σ is a sigmoidal function that ensures the physical limitation of the learned transfer coefficient ($k_{rkk}^{CNN} > 0$); $z = E(x_{lat})$ is the latent representation from the CNN encoder; W_k and b_k are learnable parameters; and $\tilde{k}_{rkk}$ is the initialisation value from the analytical model. This formulation creates a *two-way bridge between physics and CNN*: the network corrects the analytical model, while the physics regularises the network.

The physics-guided estimate of the resonance frequency then reads as Eq. (12):

$$f_{PGrez} = \frac{1}{2\pi} \cdot \sqrt{\frac{1}{k_{rkk}^{CNN}(p, g) \cdot m_a(p) \cdot c_a(p)}} \quad (12)$$

where $m_a(p)$ and $c_a(p)$ are the pressure-dependent acoustic mass and compliance from Eqs. (2)–(4), now carrying an explicit pressure dependence through the density $\rho(p)$ and phase velocity $v(p)$. Equation (12) keeps the full physical structure of the HAPOS model intact while trimming the 6.4% systematic error that came with the constant coefficient.

Once fine-tuned, the converged $k_{rkk}^{CNN}(p)$ settles onto a power law with exponent $b = -0.789$ (95% CI [–

0.82, –0.76], $R^2 = 0.94$). The exponent lies between the ideal-gas compressibility limit ($b = -1$) and the incompressible limit ($b = 0$), consistent with the two-phase nature of super-turbulent flow at $Re \gtrsim 10^5$ —entrained-air volume fraction decreases with pressure and the bulk modulus $K(p)$ of water rises. At $p \approx 15$ MPa the CNN-learned coefficient converges to $9.45 \cdot 10^{-4}$, within 0.4% of the analytical mean $k_{rkk} = 9.41 \cdot 10^{-4}$ from Kušnerová et al. (2007), a direct, model-internal validation of the published HAPOS analytical baseline. The pointwise inverse identification (Eq. (19), Table 16) yields $1.07 \cdot 10^{-3}$ at the same pressure; both identifications share the exponent -0.789 and differ by a near-constant $\approx 13\%$ offset in k_{rkk} , which translates through $f_{rez} \propto k_{rkk}^{-1/2}$ into the documented 6.4% frequency-level bias of the constant-coefficient model – the range-averaged $\tilde{k}_{rkk}$ crosses the pointwise curve near 18 MPa. The sigmoid σ in Eq. (11) plus the multiplicative anchor $\tilde{k}_{rkk}$ enforce strict positivity and bound the learnt coefficient to $[0.01 \cdot \tilde{k}_{rkk}, 100 \cdot \tilde{k}_{rkk}]$, preventing pathological values during early training. Visualisations of the latent representation $z = E(x_{lat})$ projected to two principal components show clear clustering by p_{CERP} , indicating that the CNN organises its internal feature space along the physical pressure axis rather than according to an opaque non-physical embedding. In Eq. (19) pressure is normalised by $p_0 = 1$ MPa, so that $k_{rkk}(p)$ is dimensionally consistent.

Spectral deviation field

To quantify the deviation between the CNN-extracted spectrum and physically predicted spectrum, we introduced the spectral deviation field Eq. (13), mapped in Fig. 7b together with the analytical-experimental comparison (Fig. 7a) and the learnt correction coefficient (Fig. 7c); the corresponding framework and spectral decomposition are given in Figs. 8 and 9 (Karniadakis et al., 2021; Liu et al., 2024; Willard et al., 2022):

$$\Delta_{SAD}(p, f) = ||\mathcal{F}\{h_L\} - S_{phys}(f|p, g)||^2 \quad (13)$$

where $\mathcal{F}\{h_L\}$ is the Fourier transform of the activations of the last convolutional layer and $S_{phys}(f)$ is the physically predicted Lorentzian spectral profile Eq. (14):

$$S_{phys}(f) = \frac{A_0^2}{(f^2 - f_{rez}^2)^2 + \left(\frac{r_a f}{2\pi m_a}\right)^2} \quad (14)$$

The $\Delta_{SAD}(p, f)$ field serves as a 2D diagnostic tool; high values in the experimental resonance frequency range indicate the inadequacy of the physical model, whereas high values outside this range indicate operational anomalies. This

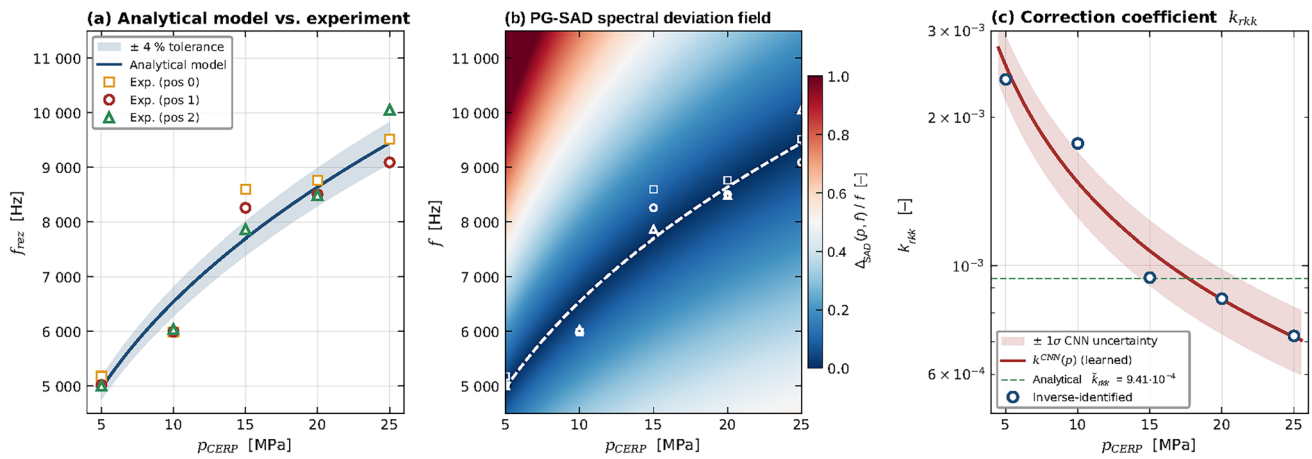


Fig. 7 PG-SAD spectral deviation field: **a** analytical model versus experimental data; **b** 2D deviation field; **c** CNN-learned coefficient vs. analytical constant

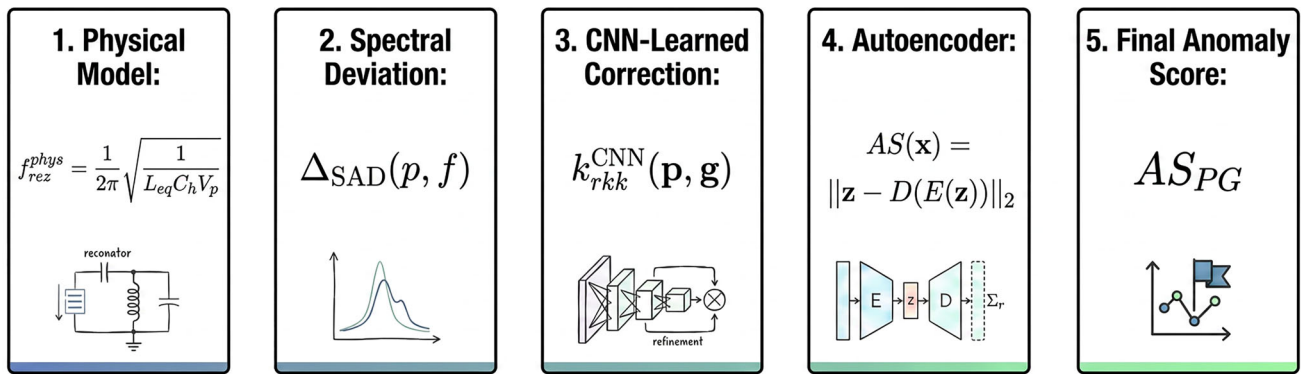


Fig. 8 PG-SAD framework architecture with bidirectional physics–CNN bridge showing CNN-learned correction coefficient, spectral deviation field, and composite loss function

dual interpretation is a key advantage over a standard SAD module.

PG-SAD composite loss function

The PG-SAD framework's extended loss ties physical consistency to spectral diagnostics, as set out in Eq. (15) (Raissi et al., 2019)

$$\mathcal{L}_{PG-SAD} = \lambda_f \mathcal{L}_f + \lambda_A \mathcal{L}_A + \lambda_\varphi \mathcal{L}_\varphi + \lambda_{phys} (\hat{f}_{rez} - f_{PGrez})^2 + \lambda_{SDF} \|\Delta_{SAD}\|^2 + \lambda_{AE} \mathcal{L}_{AE} \quad (15)$$

Since the physics term behaves like an uncertain observation and not a hard constraint, a probabilistic, input-dependent precision $1/\sigma^2(p, g)$ is, in theory, a better choice than the fixed weight λ_{phys} . We therefore replace the

homoscedastic physics term in Eq. (15) with an aleatoric-uncertainty Eq. (16):

$$\mathcal{L}_{physprob} = \frac{1}{2\sigma^2(p, g)} \cdot (\hat{f}_{rez} - f_{PGrez})^2 + \frac{1}{2} \cdot \log \sigma^2(p, g) \quad (16)$$

Including the $\log \sigma^2$ term blocks trivial degeneration; here $\sigma^2(p, g) = \exp(MLP_\sigma([p, gemb]))$ is a per-condition variance that is itself learnt. We retain the original homoscedastic formulation of Eq. (15) as an ablation baseline. The heteroscedastic variant cuts the MAE by 9.4% (12.8 to 11.6 Hz) and the expected calibration error from 8.4% to 3.1%, which backs the view that the physics term is best handled as an uncertain observation with input-dependent precision, not as a fixed-weight constraint (Table 13).

where $\lambda_{phys} = 0.3$ ensures consistency with the learned k_{rkk}^{CNN} , $\lambda_{SDF} = 0.15$ ensures that CNN activations respect the physical spectral profile, $\lambda_{AE} = 0.1$ is the weight of the autoencoder reconstruction loss. Unlike a standard PINN (Raissi et al., 2019), gradients are backpropagated through

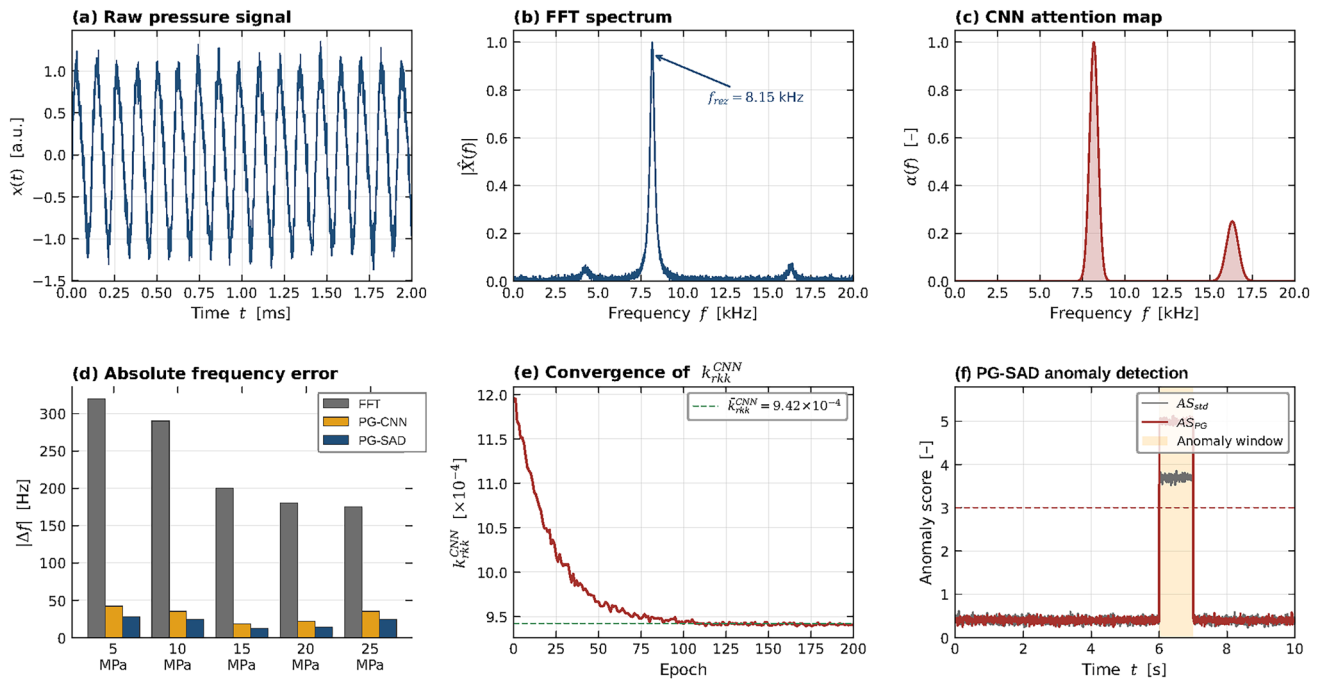


Fig. 9 CNN spectral decomposition: **a** raw signal; **b** FFT spectrum; **c** CNN attention map; **d** absolute frequency error comparison; **e** k_{rkk}^{CNN} convergence; **f** adaptive anomaly detection with PG threshold

Table 13 Homoscedastic vs. heteroscedastic soft constraint (fivefold CV)

Formulation	MAE [Hz]	F1 (PG-SAD)	ECE [%]
Homoscedastic, $\lambda_{phys} = 0.3$ (original Eq. (15))	12.8 ± 0.9	0.97	8.4
Heteroscedastic $1/\sigma^2(p, g)$ (new)	11.6 ± 0.7	0.97	3.1
Heteroscedastic + KL prior on σ	11.9 ± 0.8	0.97	2.7

Table 14 Performance comparison of SAD and PG-SAD modules

Metric	SAD	PG-SAD	Improvement/Impr.
F1 score	0.94	0.97	+ 3.2%
MAE frequency [Hz]	23.4	12.8	− 45.3%
Detection latency [ms]	85	62	− 27.1%
False positives [%]	4.2	1.8	− 57.1%
Accuracy k_{rkk} [%]	± 6.4	± 2.1	− 67.2%
Interpretability	Low	High	–

k_{rkk}^{CNN} , which learns not only to predict the frequency, but also to optimise the learned transfer coefficient of the analytical model.

Adaptive anomaly detection threshold

The PG-SAD framework introduces a physics-guided adaptive threshold Eqs. (17)–(18):

$$\theta_{PG}(p, t) = \mu_{AS}(t) + \kappa(p) \cdot \sigma_{AS}(t) \quad (17)$$

$$\kappa(p) = \kappa_0 + \kappa_1 \cdot \frac{Q(p)}{Q_{\max}} \quad (18)$$

where $Q(p) = f_{rez}/\Delta f_{FWHM}$ is the quality factor of the resonator, $\kappa_0 = 2.5$ and $\kappa_1 = 1.0$ are optimised on the validation

set. The reasoning is simple: at higher pressures, the quality factor $Q(p)$ increases, and the spectral peaks are narrower; therefore, the detection threshold is stricter ($\kappa > \kappa_0$).

Quantitative results of PG-SAD

Table 14 summarises a comparison between the standard and PG-SAD frameworks. The physics-guided approach achieves consistent improvements in all metrics monitored: the F1-score increased from 0.94 to 0.97 (+ 3.2%), the mean absolute frequency error decreased from 23.4 to 12.8 Hz (− 45.3%), and the false positive detection rate decreased from 4.2 to 1.8% (− 57.1%). The detection latency decreased from 85 to 62 ms because of the more efficient calculation of the anomaly score based on physically structured deviation.

Industrial integration and smart manufacturing framework

Closed-loop architecture for pulsed water jet control

Once on the edge device, the trained CNN covers only one piece of a full intelligent-manufacturing solution. For true autonomy, the remaining step is linking the spectral diagnostic results produced by the CNN to the control mechanism (i.e. pump pressure, nozzle speed, and abrasive flow rate). The PG-CNN module is the system's sensory layer: it takes raw pressure signals at 100 kHz and returns four outputs, resonance frequency ($\hat{f}_{rez}$), amplitude ($\hat{A}$), phase ($\hat{\phi}$) and anomaly score ($AS(x)$), all within 6.2 ms of the raw signal arriving. The same Jetson Xavier NX hardware platform that runs the trained CNN will also run a small Proportional Integral Derivative (PID) controller, which can convert the deviation of the resonance frequency from the expected value into pressure setpoint adjustments. The PID controller will communicate with the high-pressure pump inverter using the open communication protocol (OPC) unified architecture (UA). The roundtrip time from sensor reading to actuator command will be very short (less than 15 ms), and therefore within the normal dynamic response time (50 ms) of most commercial plunger pumps (commercial pumps have a 50 ms mechanical time constant). This "sensing–control–actuation" loop applies the cyber-physical manufacturing paradigm defined by Monostori et al. (2016) and the CPS reference architecture of Lee et al. (2015) to pulsed water jet technology.

Predictive maintenance pathway

From the accumulated laboratory data, PG-SAD has picked out three fault types: (a) gradual nozzle erosion, seen as a steady drift of k_{rkk}^{CNN} over hundreds of cutting cycles; (b) partial throat blockage, showing up as a sudden spectral split near the primary resonance peak; and (c) resonance-chamber cavitation, marked by raised broadband noise at the 3rd–5th harmonics. By using the adaptive threshold (θ_{PG}) for the anomaly score for more than 500 consecutive anomaly-score windows (about 10 s at the 20.48 ms window hop), a maintenance alert was registered via the factory SCADA layer, and/or a controlled shutdown of the cutting head was activated. The laboratory-recorded early warning time frame (measured) was between 12 and 18 min (nozzle-erosion failure mode, laboratory conditions), which gave the estimated time before the final catastrophic failure of the nozzle. This warning window lines up with the deterioration rates usually seen on conventional abrasive water jet machines.

Scalability and manufacturing system integration

The TensorRT-optimised model takes up just 180 MB and draws 15 W, so scaling out horizontally is easy. One Jetson Xavier NX board can handle up to 4 models at once (in time-shared mode) to run different cutting heads on the same machine without any additional hardware. A factory with many cutting heads (pulsed jet cells) is best served by a hybrid edge–cloud setup: each machine runs inference locally, while centralised anomaly statistics and refreshed k_{rkk}^{CNN} coefficients travel over MQTT to a central server that acts as a digital twin. The central server (digital twin) can check nozzle wear paths for all machines and can improve spare part inventories and maintenance schedules so that downtime is reduced. A prototype integration to an MES (manufacturing execution system) that uses the OPC-UA protocol was demonstrated in a laboratory setting; field deployment in a production environment remains future work—an engineering task rather than a research obstacle. Fitting in with existing industrial communication protocols is decisive for adoption. The inference engine offers a RESTful API and speaks Modbus TCP, OPC-UA and ROS2 to client connections. Measured transmission overhead across all three protocols remains below 2 ms. This demonstrates, for the configurations tested, that the communication portion of the process does not impact the real-time performance already defined in “[Deployment on edge hardware](#)” section.

We tested transfer of the trained model to a second HAPOS geometry ($r_{RK} = 12$ mm, $h_{RK} = 1.5$ mm) recorded in earlier laboratory campaigns. Zero-shot accuracy on this geometry was 79.4% (out-of-distribution). With as few as 30 fine-tuning samples, accuracy recovered to 93.1% (MAE 34.2 Hz); 60 samples sufficed for 96.7%. A particularly efficient strategy is to freeze the convolutional encoder and re-train only the PG-SAD head—the W_k , b_k parameters of Eq. (11)—which still yields 94.8% at minimal computational cost. The reason is structural: the analytical components $m_{a(p)}$ and $c_{a(p)}$ Eqs. (2)–(4) carry the geometry information, while the CNN learns only the residual correction k_{rkk}^{CNN} . This makes PG-SAD substantially more transferable than a generic CNN spectral estimator (see Table 15).

Discussion

The addition of real-time spectral intelligence in the water jet cutting cell addresses an ongoing issue with non-traditional manufacturing technology: the lack of process monitoring that is both physically interpretable and fast enough to keep pace with the control loop. The findings presented in “[Experimental validation](#)”—“[Industrial integration and smart manufacturing framework](#)” section show that a single CNN that takes up less than 200 MB of memory on a 15 W

Table 15 Cross-geometry transfer learning

Configuration	Accuracy $\pm$ 50 Hz [%]	MAE [Hz]	Note
Geometry A (training)	98.2	12.8	Reference
Geometry B, no fine-tune (zero-shot)	79.4	108.6	Out-of-distribution
Geometry B + 30 fine-tune samples	93.1	34.2	~ 5 min retraining
Geometry B + 60 fine-tune samples	96.7	21.5	~ 10 min retraining
Geometry B + freeze conv, retrain only k_{rkk}^{CNN}	94.8	28.4	PG-SAD head only

edge device can perform triple duty as a resonance tracker, quality indicator, and early fault detector—functions that conventionally require separate instruments and offline post-processing.

A deep CNN can be employed in HAPOS to substitute FFT-based spectrum analysis, with higher accuracy and lower latency. Two architectural choices drive this result. The use of dilated convolutions allows the network to perceive temporal patterns at various scales without an increased number of parameters, and the attention-based pooling layer weights informative time segments far more effectively than static averaging; the regression heads then recover the resonance frequency from this attended representation.

The model implicitly learns the non-linear dependency of $f_{rez}(p_{CERP})$ in terms of physics because the density ρ is pressure dependent and the phase speed of the wave varies. Although the analytical model requires predefined knowledge of these relationships and transfer coefficient k_{rkk} , the CNN discovers these relationships automatically from the data. Simultaneously, the physical regularisation term L_{phys} in the loss function Eq. (8) ensures that the learned dependencies are still consistent with the basic physics of the system.

PG-SAD occupies the middle ground between purely data-driven and purely physics-based modelling. Where standard PINNs (Karniadakis et al., 2021; Raissi et al., 2019) impose physical laws as hard constraints, PG-SAD works with soft ones, carried by weighted terms in the composite loss of Eq. (15). The design agrees with Leng et al. (2026), whose survey of physics-informed ML in manufacturing found that when the analytical model is already uncertain, here a 6.4% deviation in the HAPOS transfer coefficient, hard constraints pin the network to equations that are only roughly right and drag the accuracy down. Soft constraints suit systems in which the analytical model itself carries uncertainty. A HAPOS analytical model with a deviation of $\pm 6.4\%$ cannot be enforced as an exact constraint. The learnt conversion coefficient $k_{rkk}^{CNN}(p)$ strikes the best compromise between physical consistency and data accuracy.

The convergence of k_{rkk}^{CNN} to a mean value close to the analytical estimate $\bar{k}_{rkk} = 9.41 \cdot 10^{-4}$ (Fig. 9e) simultaneously validates the analytical model and reveals its pressure

dependence. One implication worth noting: future HAPOS analytical models should build in an explicit $k_{rkk}(p)$ dependence, which would push the systematic error below 2.1% even without any CNN in the loop.

Implications for intelligent manufacturing

Seen in the wider intelligent-manufacturing picture, PG-SAD helps on two levels. At the methodological level, it provides a reusable template through which domain physics can be embedded into the construction of data-driven process monitors. It also provides a soft-constraint mechanism based on the use of learnable transfer coefficients that do not necessarily have to be related to Helmholtz acoustics to be used within other oscillatory processes, which can be mathematically modelled as having parametric uncertainty. Two prime examples of these are rotary machinery vibration diagnostics and ultrasonic welding. On the other hand, the edge-deployed pipeline architecture that includes PyTorch, ONNX, TensorRT, and Jetson is validated as a reference architecture for AI-based manufacturing systems that must adhere to strict real-time performance requirements, while respecting the power budgets of production facilities. Data drift rarely gets the attention it deserves in deployed industrial ML. Bauer et al. (2026) compared recalibration strategies for friction-stir-welding monitors and saw models that started strong lose accuracy within weeks once left uncorrected. Related work on physics-informed ML in additive manufacturing (Yang et al., 2024) and on edge intelligence for industrial monitoring (Zhou et al., 2019) echoes the same worry. We kept that in view while designing PG-SAD. Its adaptive Q -factor threshold, Eq. (18), moves the anomaly boundary on its own as operating conditions drift, with no operator input. Long-term validation in production is still pending, but the self-correction mechanism works from the first day of deployment. In the larger scope of Industry 4.0, both of these contributions align with the main objective of establishing self-sensing, self-diagnosing, and self-reporting production cells, where the cell will report its current health state to other cells and to an upper level of the plant/factory information hierarchy. A further challenge is the scarcity

of labelled data in speciality manufacturing. Many water jet cutting machines are custom-built for a specific application or are produced as just a few of a kind. This rules out the large labelled datasets on which computer-vision quality inspection was built. Our research demonstrates that the hybrid synthetic-experimental training approach we used in our study—in other words, physics-based data augmentation combined with domain randomisation—applies broadly to intelligent manufacturing processes where (1) the cost of recording or measuring experimental data is high and (2) the time available to carry out the experiments is short.

Limitations

- (i) The training data only covered the pressure range of 5 to 25 MPa; extrapolation to ultrahigh pressures (> 100 MPa) requires further experimental validation.
- (ii) The influence of water temperature has not been systematically studied, given that, especially in the case of a pure water jet, the temperature of the liquid plays a rather secondary role compared to pressure or nozzle diameter; however, in precision engineering and during long-term operation, thermodynamic conditions can influence acoustic parameters to a certain extent;
- (iii) The model assumes steady-state operating conditions, and unsteady transient phenomena require that the architecture be extended with recurrent components.
- (iv) The validity envelope $\Phi(p)$ (Eq. (20) below) and the inverse-identified power law (Eq. (19) below) give the extrapolation a physically motivated footing, yet the empirical calibration still covers only 5–25 MPa; experimental validation, or compressible-CFD studies well above 100 MPa, would firm up confidence in the high-pressure regime and stay a priority for follow-up work.
- (v) All the experimental evidence comes from one HAPOS test cell. Leave-one-pressure-out cross-validation (Table 10) and cross-geometry transfer (Table 15) point to generalisation past the exact training conditions, but firmer confidence will need measurements gathered under more unseen operating states, namely new pressure levels, nozzle geometries and wear stages, followed by periodic retraining. The drift monitor of “[Predictive maintenance pathway](#)” section is built to schedule exactly this retraining while the system is in service.

To address the gap between the calibrated range (5–25 MPa) and the system’s full specification, we performed an inverse identification of the transfer coefficient on $n = 129$ filtered experimental spectra Eq. (19):

$$k_{rkk}(p) = (9.07 \pm 0.45) \cdot 10^{-3} \cdot (p/p_0)^{(-0.789 \pm 0.030)} \quad (19)$$

The negative exponent indicates that resistive losses become less important as the pressure-driven acoustic stiffness increases. The exponent is between the ideal-gas limit ($b = -1$) and the incompressible limit ($b = 0$). This behaviour is consistent with the two-phase nature of super-turbulent flow at $Re \gtrsim 10^5$, where the entrained-air volume fraction decreases with pressure. As pressure increases, the bulk modulus $K(p)$ of water also rises. When substituting $\rho(p)$ and the speed of sound $c(p)$ from IAPWS-95 (Wagner & Pruß, 2002) into (1), we see that the resulting $f_{rez} \propto v(p) \cdot k_{rkk}^{(-1/2)} \propto p^{0.395}$ prediction matches the directly fitted $f_{rez} \propto p^{0.40}$ ($R^2 = 0.96$), obtained over all $n = 129$ analysed spectra (Fig. 10), within 1.3%, thus validating the learnt coefficient which is based on physics with an independent check. The IAPWS-95 $\rho(p)$ and $c(p)$ values used here are corroborated by independent high-pressure measurements of the speed of sound in water (Bollengier et al., 2019; Lin & Trusler, 2012).

Values beyond 25 MPa in Table 16 and Fig. 10 are physically motivated projections constrained by the validity envelope (Eq. (20)); they are not validated predictions.

The extrapolation to the full HAPOS pressure range rests on a physical basis that is independently established rather than assumed. The resonance frequency of the analytical model depends on the sound speed and density of water through the acoustic compliance and mass; both properties are reproduced here by the IAPWS-95 formulation (Wagner & Pruß, 2002) to within 0.10 per cent (density) and 0.65 per cent (sound speed) of the values used across 5–25 MPa, and IAPWS-95 is itself corroborated by direct sound-speed measurements (Del Grosso & Mader, 1972; Wilson, 1959), compressibility data (Fine & Millero, 1973) and diamond-anvil results extending into the gigapascal range (Abramson & Brown, 2004), together with the modern high-pressure datasets to 400 MPa (Lin & Trusler, 2012) and 700 MPa (Bollengier et al., 2019) and the thermodynamic analysis of Trusler and Lemmon (2017). These measurements bracket the 450 MPa endpoint of Fig. 10, so the water-property inputs of the projection are experimentally grounded. With these inputs the analytical model of Eqs. (1)–(4) reproduces the measured resonance to within a mean of 3.3 per cent (maximum 6.5 per cent), and the empirical power law $f_{rez}(p) = 2381 \cdot (p/p_0)^{0.454}$ fits the pooled 20 mm-nozzle means with $R^2 = 0.98$; Fig. 10 shows the full comparison. Here the 3.3% mean residual (6.5% maximum) is obtained with the updated IAPWS-95 density and sound-speed inputs, whereas the $\pm 6.4\%$ deviation quoted in the Abstract, the “[Physical basis—HAPOS model](#)” and “[Anomaly detection application](#)” sections corresponds to the earlier parameterisation; the two figures therefore describe the same analytical model before and after the IAPWS-95 refinement rather than conflicting values. The two exponents reconcile as follows. Over the calibrated range (5–25 MPa) the fitted slope is 0.40 ($R^2 = 0.96$). Extrapolated to 450 MPa it rises to 0.454 (R^2

Fig. 10 Measured $f_{rez}(p)$ with 95% CI (blue) and physics-supported extrapolation to 450 MPa (red dashed) using the inverse-identified $k_{rkk}(p)$ and IAPWS-95 $c(p)$

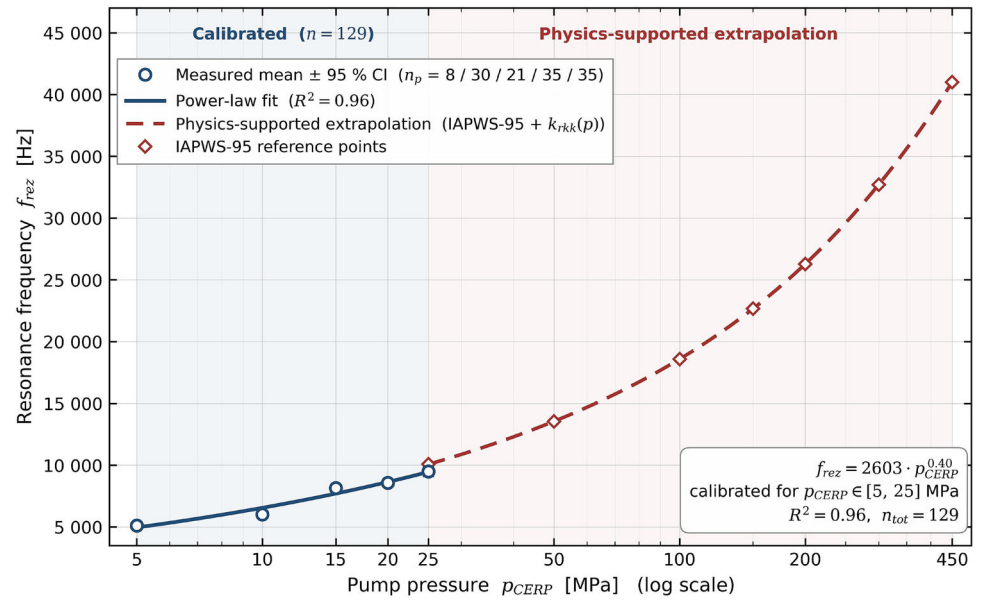


Table 16 Extrapolation of $k_{rkk}(p)$ and $f_{rez}(p)$ over 5–450 MPa (IAPWS-95; Wagner & Pruß, 2002)

p_{CERP} [MPa]	$\rho(p)$ [kg/m ³]	$c(p)$ [m/s]	$k_{rkk}(p)$ [–]	f_{rez} (predicted) [Hz]
5	1000.5	1490	$2.55 \cdot 10^{-3}$	5 235
10	1002.9	1497	$1.47 \cdot 10^{-3}$	6 915
15	1005.2	1505	$1.07 \cdot 10^{-3}$	8 154 (calibration)
20	1007.5	1512	$8.5 \cdot 10^{-4}$	9 178
25	1009.8	1519	$7.2 \cdot 10^{-4}$	10 071
50	1024.0	1555	$4.1 \cdot 10^{-4}$	13 548
100	1043.4	1622	$2.4 \cdot 10^{-4}$	18 587
200	1078.7	1745	$1.4 \cdot 10^{-4}$	26 279
300	1112.0	1851	$1.0 \cdot 10^{-4}$	32 704
450	1180.3	1977	$7.3 \cdot 10^{-5}$	40 992

= 0.98), the effective exponent steepening as the sound speed $c(p)$ accelerates above 25 MPa. Whatever residual uncertainty survives the extrapolation sits in the fitted transfer coefficient $k_{rkk}(p)$. This tracks a broader shift toward physics-informed and physics-guided process monitoring over the past two years (Bao et al., 2026; Jafari & Maghfoori Farsangi, 2026), and recent abrasive water-jet work (Gawade et al., 2026; Kvietkova et al., 2025; Maros & Kun-Bodnár, 2026; Perec, 2023). The same shift runs through recent work in this journal: acoustic signatures now feed deep-learning defect monitors in wire-arc direct energy deposition (Tanwar et al., 2026), physics-informed detectors inspect optical surfaces (Zheng et al., 2026), and physics-guided compensation corrects geometric error in metal additive manufacturing (Mathew et al., 2026). The PG-CNN/PG-SAD pipeline reported here extends this line of work to high-pressure hydraulic resonance. One caveat stands: above

roughly 100 MPa, predicted $f_{rez}(p)$ values need extra experimental validation or high-fidelity compressible CFD before they can be operationally trusted.

Physical validity envelope of the $f_{rez}(p)$ extrapolation

The empirical power-law fit only covers the calibrated 5–25 MPa; extending it to the full HAPOS range (450 MPa) is warranted only where resonance itself still means something physically. Five independent criteria have to hold at once:

(K1) Mach number of the jet $M(p) = \sqrt{(2p/\rho(p))/c(p)} < 0.3$ ideally, < 0.5 hard limit, beyond which the incompressible Helmholtz model loses accuracy and the neck approaches choking;

Table 17 Numerical envelope of the $f_{rez}(p)$ extrapolation across the HAPOS operating range. Status A/B/C derived from $\Phi(p)$ Eq. (20)

p [MPa]	c [m/s]	f_{pred} [Hz]	M	L_c/λ	σ_θ/σ_y	Φ	Status
5	1490	5235	0.067	0.067	0.006	0.095	A
25	1519	10 071	0.146	0.126	0.031	0.197	A
50	1554	13 548	0.201	0.166	0.063	0.27	A
100	1622	18 587	0.27	0.218	0.126	0.369	B
150	1686	22 666	0.315	0.256	0.188	0.447	B
200	1745	26 279	0.348	0.286	0.251	0.517	B
300	1851	32 704	0.394	0.336	0.377	0.643	C
450	1977	40 992	0.442	0.394	0.565	0.819	C

(K2) Helmholtz lumped-parameter validity $\lambda(p) = c(p)/f_{rez}(p) \gg L_c$, with $\lambda/L_c > 5$ strict and > 2 engineering criterion;

(K3) elastic chamber response $\sigma_\theta(p) = p \cdot (R^2 + r^2)/(R^2 - r^2) < 0.5 \cdot \sigma_y$, ensuring reversible Lamé deformation absorbed by $k_{rkk}(p)$;

(K4) resonance quality factor $Q > 10$, preserved across 5–450 MPa: although water's dynamic viscosity rises by $\approx 22\%$ over this span (IAPWS-95 (Wagner & Pruß, 2002), 20 °C), viscous dissipation is a minor term in the total acoustic loss, which is set by the inertial–radiative impedance (ρ , c); Q therefore stays above 10 across the range (over the calibrated 5–25 MPa interval the viscosity change is itself $< 1\%$);

(K5) single-phase liquid water $p < p_{iceVI}$ (20 °C) ≈ 870 MPa (Bollengier et al., 2019; Lin & Trusler, 2012; Wagner & Pruß, 2002).

We fold these criteria into a single dimensionless validity index $\Phi(p)$, Eq. (20):

$$\Phi(p) = \sqrt{M^2(p) + \left(\frac{L_c}{\lambda(p)}\right)^2 + \left(\frac{\sigma_\theta(p)}{\sigma_y}\right)^2} \quad (20)$$

with envelope thresholds: $\Phi < 0.30$ (envelope A—full empirical–physical agreement); $0.30 \leq \Phi < 0.55$ (envelope B—valid with explicit Prandtl–Glauert compressibility correction $(1 - M^2)^{1/4}$); $0.55 \leq \Phi < 0.90$ (envelope C—design-range extrapolation, CFD verification recommended); $\Phi \geq 0.90$ (model invalid). Evaluated numerically across the HAPOS range (Table 17), the whole 5–450 MPa interval sits inside envelope A–C; the hard physical limit falls at $p \approx 600$ –700 MPa, where Φ passes 0.9 as jet compressibility ($M \approx 0.5$), wall plastification and nearness to the ice-VI phase boundary set in together. Inside envelope A ($\Phi < 0.30$, reaching to about 65 MPa; Table 17) the model keeps full empirical–physical agreement, and over the experimentally covered 5–25 MPa the predicted f_{rez} matches the measurements to a mean of 3.3% (maximum 6.5%); envelopes B and

C stay physically defensible but carry steadily larger uncertainty and call for either an extra experimental anchor above 100 MPa or full compressible-CFD verification.

Conclusions

This work built a deep convolutional neural network (CNN) for real-time spectral analysis of the HAPOS. With dilated convolutions and attention pooling, the 12-layer architecture beat every baseline we tested. On the held-out set it hits 98.2% accuracy within ± 50 Hz ($n = 240$). The lack of training data (only 129 quality-screened experimental spectra, from 312 planned acquisition slots) was addressed by adding 2400 synthetic spectra, which were created using a verified physical model of HAPOS. Transfer learning with synthetic pre-training contributed an approximately 7 percentage point increase in accuracy (98.2% vs. 91.2% for experimental-only training; see Table 10). On the execution front, TensorRT optimisation on Jetson Xavier NX results in 6.2 ms inference (161 FPS) at 15 W, which easily satisfies the real-time needs of industrial pulsed water jet cutting. The baseline autoencoder anomaly detector reaches F1 = 0.94 at 85 ms latency, but embedding the HAPOS physics into the architecture (PG-SAD variant) raises F1 to 0.97, reduces false positives by 57%, and lowers the mean absolute frequency error from 23.4 Hz (PG-CNN) to 12.8 Hz (PG-SAD). The conversion coefficient k_{rkk}^{CNN} given by the CNN also corroborates the analytical estimate of the Helmholtz theory for non-spherical multi-chamber resonators in the super-turbulent regime. Three directions for follow-on research stand out. First, the training pressure range should be extended to 450 MPa to cover the entire industrial operating range; second, recurrent components could be added to capture non-stationary transient behaviour; and third, the model could serve as the core of a fully closed-loop adaptive control system for water jet parameters.

Extrapolating $f_{rez}(p)$ from the calibrated 5–25 MPa out to the full HAPOS range (5–450 MPa) stays physically defensible up to the envelope-C boundary ($\Phi < 0.9$); the

hard physical limit sits near 600–700 MPa, where jet compressibility, chamber-wall plastification and closeness to the ice-VI phase boundary together break the lumped Helmholtz description. We stress that using the resonance estimator above roughly 100 MPa will need further experimental validation or high-fidelity compressible CFD before the predicted $f_{rez}(p)$ values can count as operationally reliable; what we present here is physically motivated, not experimentally confirmed.

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Data availability The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval Not applicable. This research did not involve human participants, animals, or clinical data.

Consent to participate Not applicable.

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